

# Victor Chua

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Singaporean Citizen | [linkedin.com/in/vicchua123](https://www.linkedin.com/in/vicchua123) | Internship: May 11 – August 11 2027

Generative AI & Multimodal Vision Researcher | World Models | PyTorch & Fine-Tuning

## Summary

Computer Science undergraduate at Nanyang Technological University (Turing AI Scholars Programme) specializing in multimodal generative AI, video understanding, and computer vision model fine-tuning. Creator of a fine-tuned vision model published on HuggingFace and developer of multimodal video analysis pipelines using TwelveLabs. Experienced across 3 AI engineering internships in PyTorch, dataset curation, and interactive agentic architectures.

## Technical Skills

### Generative AI & Multimodal Vision

- **PyTorch**, **TwelveLabs**, **MobileNetV3**, Vision-Language Models, Multimodal AI, Transfer Learning
- Optical Music Recognition, Image & Video Processing, HuggingFace, Failure Mode Analysis
- **QLoRA** / **PEFT**, 4-bit Quantization, Feature Extraction, Dataset Curation, Data Augmentation

### Agentic & LLM Architecture

- **LangGraph**, **RAG**, **ChromaDB**, Interactive Agents, State Machines, Tool-Calling, Prompt Engineering

### Programming & ML Frameworks

- **Python**, **TensorFlow**, **Scikit-learn**, **SQL**, **JavaScript/TypeScript**, **Dart**, **FastAPI**, **Node.js**

### Infrastructure & Cloud

- **Linux**, **Git**, **AWS** (S3, Cost Explorer), **Firebase**, **Elasticsearch**, **SQLite**, **Colab GPU Training**

### Data Systems & Engineering

- **Grafana**, **Kafka**, **RabbitMQ**, **Amazon S3**, **Data Pipelines**, **Automated Monitoring**, **Performance Profiling**

## Work & Research Experience

### Published Multimodal Vision Model & Custom Dataset Pipeline

Jun 2026 – Present

*Creator & Developer, Drum Hub (AI Desktop Application)*

- Trained a dual-head MobileNetV3 Optical Music Recognition model via two-phase transfer learning on a custom dataset; evaluated and published the trained vision model weights on **HuggingFace**.
- Curated an original labelled vision dataset from scratch using synthetic data generation, automated annotation, and image augmentations to train custom deep learning models.
- Designed a mixed-initiative **Agentic AI** architecture using **LangGraph** with explicit state machines, tool-calling schemas, and conversation memory to process complex user inputs reliably.

### Multimodal Video Understanding & On-Premises LLM Inference

May 2026 – Aug 2026

*AI & Software Engineer Intern, SUSS Digital Solutions & AI*

- Automated document and video content categorization using multimodal analysis (**TwelveLabs**) and structured LLM prompts, accelerating internal review workflows across multiple university departments.
- Built an on-premises **LLM** inference pipeline for privacy-sensitive university documents, enforcing strict data confidentiality while eliminating manual administrative data processing delays.

### Generative Interactive Interface Research

May 2025 – Aug 2025

*Research Assistant, NTU Turing AI Scholars Programme (TAISP)*

- Engineered bidirectional generation logic for forward and backward LLM text suggestions, enabling real-time human-in-the-loop interactive translation workflows across complex datasets.

### Production RAG Pipeline & Context Vector Search

Dec 2025 – Jan 2026

*Machine Learning Intern, Yap Motor Singapore*

- Architected and deployed a production **RAG** system leveraging Google Gemini and **ChromaDB** vector embeddings to deliver context-aware automotive diagnostic recommendations to 24 mechanics daily.

### Enterprise Data Pipeline Monitoring & Bottleneck Optimization

May 2023 – May 2024

*Data Engineer Intern, DBS Bank Singapore*

- Monitored large-scale data pipeline processing **3,000+** jobs daily; developed Grafana dashboards that identified bottlenecks, enabling engineering teams to achieve **20%** RAM reduction and **15%** faster execution.

## Education

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**Nanyang Technological University (NTU)**

Aug 2024 – Dec 2028

Bachelor of Computing (Computer Science) with Turing AI Scholars Programme (TAISP)

- Advanced Coursework: AI Fundamentals, Calculus, Linear Algebra, Probability & Statistics

## Leadership & Activities

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**TAISP x MLDA Workshop Lead**, NTU School of Computing

2025

Led a Linear Regression and Neural Networks workshop for **207 participants**; designed end-to-end coding exercises covering practical ML pipelines.